

Oswald Gyabaah

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PROFESSIONAL SUMMARY

Senior Infrastructure Engineer with 5+ years building and operating distributed systems and infrastructure tooling at scale. Architected high-availability Kubernetes clusters handling 3M+ daily requests at 99.97% uptime across multi-cloud environments. Deep expertise in fleet management, infrastructure control plane operations, debugging complex distributed systems, and API-driven infrastructure automation. Built backend platforms processing 1M+ daily jobs and PaaS control planes from scratch. Strong foundation in Python and systems programming with growing proficiency in Go and Rust. Currently developing AI agents for production infrastructure automation.

CORE COMPETENCIES

Distributed Systems at Scale

Infrastructure Tooling & Automation

Fleet Management & Control Plane Ops

API Design & System Integration

Kubernetes & Container Orchestration

Debugging & Root-Cause Analysis

High-Availability Architecture

Performance Optimization

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD May 2025 - Present

Manager, Finacle Services Team | Accra, Ghana

- Manage **800+ production pods** on Tanzu Kubernetes Grid running on-prem Core Banking System, overseeing full cluster lifecycle: deployments, patching, and incident response for **1.5M+ daily banking transactions**
- Lead L2 support operations, correlating transactional and application logs across interfaces and core banking microservices via the **Elastic stack**, driving systematic root-cause analysis and cross-functional remediation
- Manage observability stack with dashboards tracking CPU/memory trends, transaction volumes, response times, and categorized exceptions - enabling threshold-based alerting and proactive scaling
- Managed end-to-end QA for core banking migration, leading a team executing **10,000+ test cases** across 13 modules and 33 internal applications
- Designed and deployed a bank-wide second factor verification service securing core banking logins for 600+ internal users

Guaranty Trust Bank Ghana LTD Apr 2024 - Present

DevOps Engineering Lead | Accra, Ghana

- Architected **high-availability Kubernetes clusters** across multiple on-prem environments handling **3M+ daily requests with 99.97% uptime SLA**; implemented auto-scaling and self-healing mechanisms
- Founded the bank's DevOps team, leading engineers to establish modern delivery practices - Kubernetes, containerized deployments, and automated release workflows
- Automated **95% of infrastructure configuration management** using Ansible, Python, and Bash - transforming multi-day manual processes into reproducible, version-controlled deployments
- Implemented CI/CD automation cutting deployment time by ~90% and increasing release velocity ~5x; introduced GitOps with FluxCD for consistent deployments
- Introduced unified observability stack (Prometheus, Grafana, Elastic Stack, Sentry) for centralized logging, tracing, and real-time alerting
- Manage **multi-cloud infrastructure** across Azure and AWS: resource provisioning, private VPC links to on-prem systems, access control, cost optimization

Chief Technical Officer

- Architected and built full platform end-to-end: requirements gathering, **API architecture design**, database modeling, microservices implementation on AWS (EKS/ECS) with autoscaling and inter-service communication
- Developed **high-throughput background processing system** handling 1M+ daily jobs (payments, messaging, notifications) with failover and provider fallback integrations
- Reduced peak latency by **90%** through systematic database optimizations: query tuning, caching layers, and indexing strategies
- Designed **disaster recovery architecture**: parallel GCP production mirroring AWS using GKE, Cloud SQL, and Cloud Storage with dual-environment CI/CD
- Automated infrastructure provisioning with Terraform, reducing setup time by 90%

PROJECTS

Virk Cloud Infrastructure Control Plane

Built a production-grade Kubernetes PaaS with an infrastructure control plane for fleet management - enabling startups to deploy clusters instantly without operational overhead. API-driven configuration management, automated cluster lifecycle, and integrated observability on bare-metal VPS. RKE2, FluxCD, Terraform, Prometheus, Grafana, API-driven orchestration

AI Agents for Infrastructure Operations Current

Building specialized AI agents for QA automation and sysadmin operations on production Linux servers. Developing agent-driven tooling leveraging RAG, tool-use, and multi-agent orchestration patterns. Python, LLM APIs, RAG, Multi-agent orchestration

Education Research and Innovation Foundry Research

Leading AI-driven research initiatives in adaptive learning, cognitive modeling, and educational analytics. Python, ML/AI, LLMs

EDUCATION

BSc Computer Engineering - Ashesi University
Berekuso, Ghana

Sep 2016 - May 2020

CERTIFICATIONS

Certified Oracle DevOps Professional - Oracle
Certified Oracle Cloud Solutions Architect Associate - Oracle
AI Engineering Specialization (Agents, RAG, LLM Applications) - ByteByteAI

SKILLS

Languages: Python, Bash, Go (learning), Rust (learning) **Cloud & Infrastructure:** AWS, Azure, GCP, OCI, VMware, Bare Metal
Container Orchestration: Kubernetes (EKS, GKE, TKG, RKE2), Docker, Helm, Istio
CI/CD & GitOps: GitHub Actions, FluxCD, ArgoCD, Azure DevOps, Terraform, Ansible **Observability:** Prometheus, Grafana, ELK Stack, Sentry
Architecture: Distributed Systems, API Design, Microservices, Disaster Recovery, Multi-cloud
Networking & Security: IPSec/OpenVPN, TLS/SSL, WAF, OAuth2, Data Sovereignty
AI/Agents: LLM Integration, RAG Systems, AI Agent Development